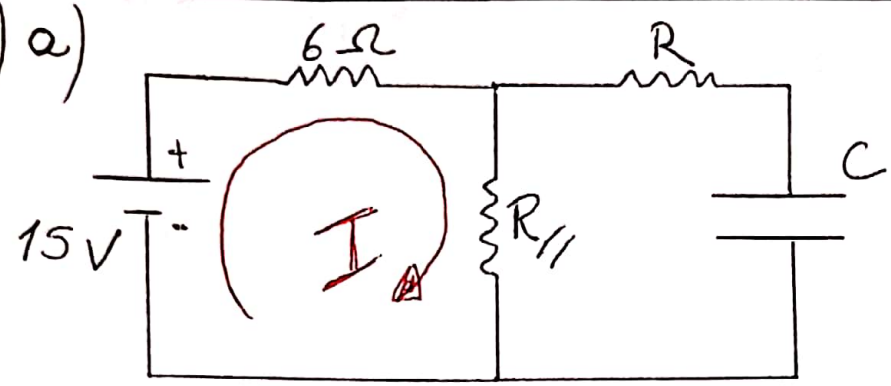
1) a)



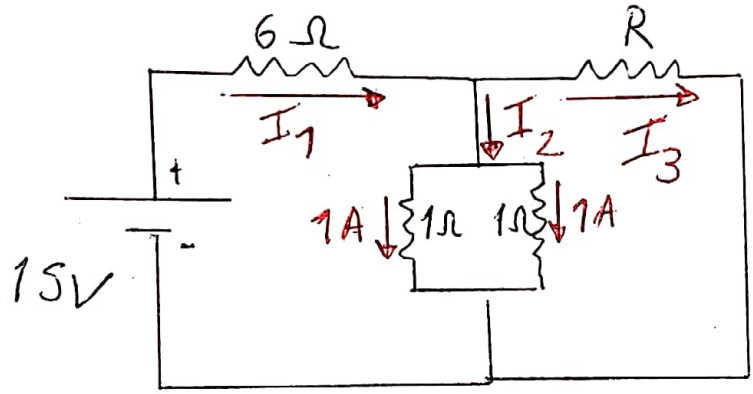
$\frac{1}{R_{//}} = \frac{1}{1\Omega} \times 2 + \frac{1}{2\Omega}$

$\rightarrow R_{//} = 0{,}4\,\Omega$

$0 = 15V - 6{,}4\,\Omega\, I \rightarrow I = 2{,}34\,A$

$V_{CAP} = R_{//} I = \frac{15}{16} V = \frac{Q}{200\mu F} \rightarrow \boxed{Q = 187{,}5\,\mu C}$

b)



$\frac{1}{R_{//}} = \frac{1}{1\Omega} \times 2 \rightarrow R_{//} = 0{,}5\,\Omega$

$I_2 = 1A + 1A \rightarrow I_2 = 2\,A$

$0 = 15V - 6\Omega I_1 - 0{,}5\Omega \times 2A$

$\rightarrow I_1 = \frac{14}{6} A \rightarrow I_1 = \frac{7}{3} A$

$I_3 = I_1 - I_2 \rightarrow I_3 = \frac{1}{3} A$; $R I_3 = 1V \rightarrow \boxed{R = 3\,\Omega}$

c) $P_{6\Omega} = 6\Omega \cdot \left(\frac{7}{3} A\right)^2 = 32{,}67\,W = 7{,}815\,\frac{CAL}{s}$

$Q = 7{,}815\,\frac{CAL}{s} \times 4 \times 60\,s = 1875{,}6\,CAL = 1\,\frac{CAL}{g°C} \times 50\,g\,\Delta T$

$\rightarrow \boxed{\Delta T = 37{,}5\,°C}$

2) a) $T_1 = 373\,K$; $T_2 = 273\,K$; $Q_A = 80\,J$; $\varepsilon_{CAR} = 1 - \frac{273}{373} = 0{,}268$

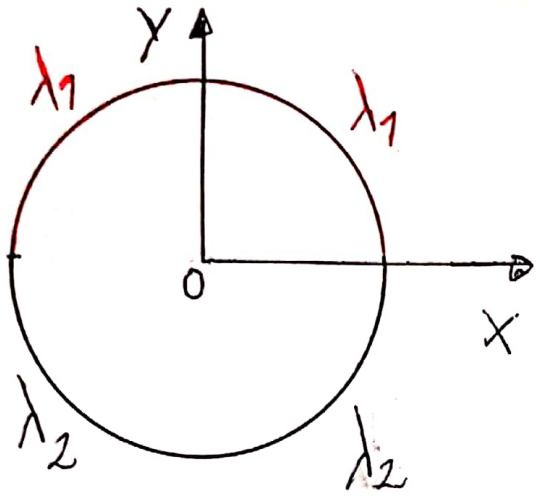
$\frac{Q_C}{80J} = -\frac{273}{373} \rightarrow \boxed{Q_C = -58{,}55\,J}$; $0 = Q_A + Q_C - W_{ciclo} \rightarrow \boxed{W_{ciclo} = 21{,}45\,J}$

b) $\varepsilon_R = \frac{1}{2}\varepsilon_{CAR} \rightarrow \varepsilon_R = 0{,}134 = \frac{W}{Q_A} = \frac{15\,J}{Q_A} \rightarrow Q_A = 111{,}9\,J$

$\varepsilon_R = 1 + \frac{Q_C}{Q_A} \rightarrow 0{,}134 = 1 + \frac{Q_C}{111{,}9\,J} \rightarrow \boxed{Q_C = -96{,}9\,J}$

Escaneado con CamScanner

3)



$$\vec{E}_0 = \vec{E}_1 + \vec{E}_2 =$$

$$= \frac{k\lambda_1}{R}(\cos 180 - \cos 0)\hat{j} + \frac{k\lambda_2}{R}(\cos 360 - \cos 180)\hat{j}$$

$$\vec{E}_0 = \frac{k}{R}(-2)\lambda_1\hat{j} + \frac{k}{R}2\lambda_2\hat{j} \rightarrow \vec{E}_0 = \frac{2k}{R}(\lambda_2 - \lambda_1)\hat{j}$$

$$|\vec{E}_0| = \frac{2k}{R}|\lambda_2 - \lambda_1| = \frac{2k}{R}\lambda_1 \rightarrow \lambda_2 - \lambda_1 = \lambda_1 \text{ o } \lambda_1 - \lambda_2 = \lambda_1 \rightarrow$$

$$\rightarrow \boxed{\lambda_2 = 2\lambda_1 \text{ o } \lambda_2 = 0} \rightarrow \lambda_2 = 14\,\mu C/m \text{ o } \lambda_2 = 0$$

4) DENTRO DEL CILINDRO EL CAMPO ES NULO $\rightarrow \boxed{W = 0}$

5) a) $\vec{F} = q\vec{E} = kq\lambda\left[\frac{1}{x_3 - x_2} - \frac{1}{x_3 - x_1}\right]\hat{i} \rightarrow$

$$\rightarrow \vec{F} = 9\times10^9\,\frac{Nm^2}{C^2} \times 8\times10^{-6}\,C \times 20\times10^{-6}\,\frac{C}{m}\left[\frac{1}{0{,}22m - 0{,}14m} - \frac{1}{0{,}22m - 0{,}06m}\right]\hat{i}$$

$$\rightarrow \boxed{\vec{F} = 9\,N\,\hat{i}}$$

b) $dV = \frac{k\delta q}{r} = \frac{k\lambda dx'}{x - x'} \rightarrow V = k\lambda\int_{x_1}^{x_2}\frac{dx'}{x - x'} = k\lambda(-1)\ln(x - x')\Big|_{x_1}^{x_2}$

$$\rightarrow V = k\lambda\left[\ln(x - x_1) - \ln(x - x_2)\right] \rightarrow V = k\lambda\ln\left(\frac{x - x_1}{x - x_2}\right)$$

$$W = -q\Delta V = -kq\lambda\left[\ln\left(\frac{x_4 - x_1}{x_4 - x_2}\right) - \ln\left(\frac{x_3 - x_1}{x_3 - x_2}\right)\right]$$

$$W = kq\lambda\ln\left[\frac{(x_3 - x_1)(x_4 - x_2)}{(x_3 - x_2)(x_4 - x_1)}\right] = 9\times10^9\,\frac{Nm^2}{C^2} \times 8\times10^{-6}\,C \times 20\times10^{-6}\,\frac{C}{m}\ln\left(\frac{16\times12}{8\times20}\right)$$

$$\boxed{W = 0{,}263\,J}$$

Escaneado con CamScanner